

**Multiple Choice:**

1. If the output impedance of an amplifier is Z_{out} without feedback, then with negative voltage feedback, its value will be _____.
A. $\frac{Z_{out}}{1 + A\beta}$
B. $Z_{out} (1 + A\beta)$
C. $\frac{1 + A\beta}{Z_{out}}$
D. $Z_{out} (1 - A\beta)$
2. In a phase-shift oscillator, we generally use _____ RC sections.
A. three
B. two
C. four
D. None of the above
3. The gain of an amplifier with feedback is known as _____ gain.
A. closed loop
B. open loop
C. resonant
D. None of the above
4. In a Colpitt's oscillator, feedback is obtained _____.
A. by magnetic induction
B. by a tickler coil
C. from the center of split capacitors
D. None of the above
5. An oscillator employs _____ feedback.
A. positive
B. negative
C. both positive and negative
D. None of the above
6. The basic purpose of applying negative feedback to an amplifier is to _____.
A. increase gain
B. reduce distortion
C. keep the temperature within limits
D. None of the above
7. If the stage gain is 20 without feedback and 12 with negative feedback, then what is the feedback factor?
A. 0.033
B. 3/5
C. 5/3
D. 1/5
8. When voltage feedback (negative) is applied to an amplifier, its input impedance _____.
A. is decreased
B. is increased
C. becomes zero
D. remains the same
9. An amplifier without feedback delivers an output voltage of 25V with 6% second harmonic distortion when the input voltage is 20 mV. Negative feedback is applied to the amplifier so that an output voltage of 25V with 0.1% distortion is obtained. Calculate the required feedback factor.
A. 0.05
B. 0.03
C. 0.01
D. 0.08
10. In order to obtain good gain stability in a negative feedback amplifier _____.
A. $A\beta = 1$
B. $A\beta < 1$
C. $A\beta \gg 1$
D. None of the above
11. If you move towards an oscillating circuit, its frequency changes because of _____.
A. hand capacitance
B. noise of foot
C. movement of body
D. None of the above
12. An oscillator produces _____ oscillations.
A. damped
B. undamped
C. modulated
D. None of the above
13. If you move towards an oscillating circuit, its frequency changes because of _____.
A. hand capacitance
B. noise of foot
C. movement of body
D. None of the above
14. In a phase-shift oscillator, we generally use _____ RC sections.
A. three
B. four
C. two
D. five
15. An oscillator converts _____.
A. a.c. power into d.c. power
B. d.c. power into a.c. power
C. mechanical power into a.c. power
D. electrical power into a.c. power
16. Series current negative feedback occurs when the feedback voltage is proportional to the _____.
A. output voltage
B. output current
C. output impedance
D. None of the above
17. When current feedback (negative) is applied to an amplifier, its input impedance _____.
A. remains the same
B. is increased
C. is decreased
D. None of the above
18. Determine the voltage gain with feedback for a voltage-series feedback having $A = -100$, $R_i = 15 \text{ k}\Omega$, $R_o = 20 \text{ k}\Omega$, and a feedback of $\beta = -0.25$.
A. -3.85
B. -5.38
C. -8.35
D. -3.58
19. In an emitter follower, we employ _____ negative feedback.
A. 50 %
B. 25 %
C. 75 %
D. 100 %



20. The frequency stability of the oscillator output is maximum in _____.
- A. LC oscillators
B. phase-shift oscillators
C. crystal oscillators
D. Wien bridge oscillator
21. When current feedback (negative) is applied to an amplifier, its output impedance _____.
- A. is increased
B. is decreased
C. remains the same
D. None of the above
22. A Wien bridge oscillator uses _____ feedback.
- A. positive
B. negative
C. both positive and negative
D. None of the above
23. In the expression for gain with negative feedback, the term $1 + \beta A$ is known as _____.
- A. gain factor
B. feedback factor
C. sacrifice factor
D. None of the above
24. Hartley oscillator is commonly used in _____.
- A. radio receivers
B. radio transmitters
C. TV receivers
D. None of the above
25. In a transistor oscillator, the active device is _____.
- A. LC tank circuit
B. biasing circuit
C. transistor
D. None of the above
26. A 1 mH inductor is available. Choose the capacitor values in a Colpitts oscillator so that $f = 1$ MHz and $\beta = 0.25$.
- A. 121.5 pF, 31.6 pF
B. 126.5 pF, 30.1 pF
C. 126.5 pF, 31.6 pF
D. 125.5 pF, 32.6 pF
27. In an LC oscillator, if the value of L is increased four times, then frequency of oscillations is _____.
- A. decreased 2 times
B. increased 2 times
C. decreased 4 times
D. increased 4 times
28. In an LC oscillator, the frequency of oscillations is _____ L or C.
- A. inversely proportional to square root of
B. directly proportional to
C. independent of the values of
D. proportional to square of
29. If the non-linear distortion in an amplifier is D without feedback, then with negative voltage feedback it will be _____.
- A. $\frac{D}{1 + A\beta}$
B. $D(1 + A\beta)$
C. $\frac{1 + A\beta}{D}$
D. $D(1 - A\beta)$
30. In an LC circuit, when the capacitor energy is maximum, the inductor energy is _____.
- A. maximum
B. minimum
C. half-way between maximum and minimum
D. None of the above
31. In a phase-shift oscillator, the frequency determining element are _____.
- A. L and C
B. R and C
C. R, L and L
D. None of the above
32. Determine the input impedance with feedback for a voltage-series feedback having $A = -100$, $R_1 = 15$ k Ω , $R_o = 20$ k Ω , and a feedback of $\beta = -0.25$.
- A. 290 Ω
B. 190 Ω
C. 390 Ω
D. 490 Ω
33. In a phase-shift oscillator, 180° phase-shift is obtained by _____.
- A. a transformer
B. three RC sections
C. LC tank circuit
D. none of the above
34. The operating frequency of a Wien-bridge oscillator is given by _____.
- A. $\frac{1}{2\pi\sqrt{LC}}$
B. $\frac{1}{2\pi RC}$
C. $\frac{1}{4\pi LC}$
D. $\frac{1}{29RC}$
35. If the input impedance of an amplifier is Z_{in} without feedback, then with negative voltage feedback, its value will be _____.
- A. $\frac{Z_{in}}{1 + A\beta}$
B. $Z_{in}(1 + A\beta)$
C. $\frac{1 + A\beta}{Z_{in}}$
D. $Z_{in}(1 - A\beta)$
36. Feedback circuit _____ frequency.
- A. is independent of
B. is strongly dependent on
C. is moderately dependent on
D. none of the above
37. In an LC oscillator, the frequency of oscillations is given by _____.
- A. $\frac{1}{2\pi\sqrt{LC}}$



- B. $\frac{2\pi}{\sqrt{LC}}$
C. $\frac{\sqrt{LC}}{2\pi}$
D. $\frac{2\pi}{\sqrt{LC}}$
38. If the output of an amplifier is 10 V and 100 mV from the output is fed back to the input, then what is feedback factor?
- A. 10
B. 0.1
C. 0.01
D. 0.15
39. Determine the output impedance with feedback for a voltage-series feedback having $A = -100$, $R_1 = 15 \text{ k}\Omega$, $R_o = 20 \text{ k}\Omega$, and a feedback of $\beta = -0.25$.
- A. 769Ω
B. 880Ω
C. 669Ω
D. 445Ω
40. When negative feedback is applied to an amplifier, its gain _____.
- A. is increased
B. is reduced
C. remains the same
D. none of the above
41. If A and A_f are the gains of an amplifier without feedback and with negative feedback, then feedback fraction is given by _____.
- A. $\frac{1}{A} - \frac{1}{A_f}$
B. $\frac{1}{A} + \frac{1}{A_f}$
C. $\frac{1}{A_f} + \frac{1}{A}$
D. $\frac{1}{A_f} - \frac{1}{A}$
42. In an LC oscillator, if the value of L is increased four times, then frequency of oscillations is
- A. decreased 2 times
B. increased 2 times
C. decreased 4 times
D. increased 4 times
43. Quartz crystal is most commonly used in crystal oscillators because _____.
- A. it is easily available
B. it has superior electrical properties
C. it is quite inexpensive
D. None of the above
44. If the feedback factor in an amplifier is 0.01, then gain with negative feedback is approximately _____.
- A. 10
B. 1000
C. 100
D. 500
45. The Q of crystal is of the order of _____.
- A. 100
B. 1000
C. 50
D. more than 10,000
46. The approximate operating frequency of a phase-shift oscillator is given by _____.
- A. $\frac{1}{2\pi\sqrt{LC}}$
B. $\frac{1}{2\pi RC}$
C. $\frac{1}{2\pi RC\sqrt{6}}$
D. $\frac{1}{29RC}$
47. The piezoelectric effect in a crystal is _____.
- A. a voltage developed because of mechanical stress
B. a change in resistance because of temperature
C. a change of frequency because of temperature
D. None of the above
48. A feedback circuit usually employs _____ network.
- A. resistive
B. capacitive
C. inductive
D. None of the above
49. In order to produce frequencies in the microwave region (2000-30,000 MHz), one would use _____ oscillator.
- A. Wien bridge
B. crystal
C. Hartley
D. klystron
50. The value of negative feedback fraction is always _____.
- A. less than 1
B. more than 1
C. equal to 1
D. None of the above
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